NUS National University of Singapore

IT5006 PROJECT AY2026/27 Semester 1

Keeping Olist's Delivery Promises

Predicting delivery time and late risk to set sharper delivery dates and rescue at-risk parcels before customers notice

GROUP 6

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EXECUTIVE SUMMARY

For Olist's delivery operations, we built a control center to set better delivery promises and rescue at-risk parcels

Context
Why late delivery

Olist promises a delivery date at checkout, yet **6.8% of delivered orders miss it**. We chose this problem because late orders earn 1–2★ reviews 6.7× as often (62% vs 9%) and drive 32% of all bad reviews, and 85% of variation arises after carrier handover, early enough to act.

Deliverable
Delivery control center

A delivery control center for Olist operations with two applications (dashboard/chatbot let staff act on both):

- **Apply a revised promised date** at checkout, and
- **Rescue at-risk parcels** after carrier handover, including early warnings to customers

Analytic approach
Regression + classification

- **Regression** predicts delivery days (average error on validation: 5.8 days at checkout, 5.1 at handover): it sets the revised promise with a regional buffer and estimates days remaining.
- **Classification** scores late risk, blended with a simple remaining-slack rule to rank the daily queue.

Business impact
Estimates and call-outs*

- Promises **2.8 days shorter with ~20% fewer late orders**
- Late parcels found with **6.7 days** left on average vs 2.0 by deadline
- **~R$0.46M a year** of late-parcel value reached while still rescuable

Olist public dataset, 2016–18. Test = orders from 26 May 2018. Results: Phase 2 report and Phase 3 workstreams.

* delivery speeds drift, so we reset the buffer monthly and switch models only when proven.

THE WHY · BUSINESS OPPORTUNITY

Delivery makes or breaks the e-commerce experience: Olist can protect ~R$0.46M a year and promise 2.8 days sooner

Delivery is a make-or-break moment in e-commerce...

39%
of shoppers are less likely to buy again after a late delivery

85%
expect to be told immediately about a delay

10–12%
conversion lift from precise checkout dates in retailer cases

...and Olist's customers say the same...

32%
of all 1–2★ reviews come from the 6.8% of orders that arrive late

6.7×
as likely to get a 1–2★ review when late (62% vs 9%)

2.3★ vs 4.3★
average review, late vs on-time orders

...opening an opportunity to strengthen Olist's customer experience through proactive late-delivery management

~R$0.46M*
GMV a year in ~2,400 late parcels flagged for rescue

-2.8 days*
shorter promises with ~20% fewer late orders

6.7 days*
average warning on late parcels, vs 2.0 by deadline

** Estimated potential impact*

1: HubBox 2024, Sendcloud 2025, Narvar and Shipup case studies. 2–3: Olist data 2016–18; test = May–Aug 2018 (promises 19.3 vs 22.1 days, 2.8% vs 3.5% late). R$: Sep 2017–Aug 2018 item value (excl. freight) of still-rescuable late parcels x test catch rate (58%)

HOW THE CONTROL CENTER WORKS

Two applications along an order's journey

Checkout | Payment approved | Seller packs | Carrier handover | In transit | Delivered

Apply a revised promised date — At checkout

What it does: shows each buyer a delivery date set from our predicted delivery time plus a safety buffer for their region.

In operation: every order gets the revised date at checkout. Each month the buffer is reset from recent deliveries, so dates keep pace with real delivery speed.

Owner: commercial and CX set the promise policy; the data team runs the monthly reset.

Rescue at-risk parcels — From carrier handover

What it does: scores each parcel's risk of missing its date, from our risk model and the spare time left on its route.

In operation: one ranked queue each morning. Expected days remaining tell staff whether to ask the carrier to expedite.

Owner: logistics ops, who set the daily list size.

↳ Pre-empt customer expectations — Customer service

Message customers whose parcel will likely miss its date, with a new expected date. If it does arrive late, send a voucher; sellers follow up when their delay caused it.

Behind both · data team: weekly drift checks, monthly buffer reset, shadow retraining

LIMITATIONS, LEARNINGS AND NEXT STEPS

What to watch, what we learned, and where to go next

Model limitations

- Seasonal peaks: the monthly update reacts too late to a Black Friday surge
- The risk score ranks parcels well, but its % values run about twice too high
- No carrier tracking or records of actions, so we can't yet prove how many parcels a rescue saves

What else we tried
but didn't implement

- Retraining: On later orders, weekly retraining using all available history reduced average checkout error from 5.0 to 4.8 days. The monthly model trained on recent orders performed best in earlier validation, but its error averaged 5.8 days on the same later orders. We kept the existing model.
- Risk model alone: fell behind a remaining-slack rule when conditions shifted; blending both matched the rule and gave earlier warning
- Stronger regularization slightly improved validation error, but worsened later-period error
- Fixing only promises that look badly off: lost most of the gain of updating every promise

Data drift

- Delivery time fell from 13.3 days in training to 8.8 days in the later test period. Of the 4.5 days saved, 3.8 came after carrier handover.
- Promised dates also shortened. Customers received an average delivery promise of 27.8 days in June 2018 and 15.2 days in August.
- Proposed response: Monitor prediction errors weekly. Test monthly model updates alongside the current model on the same new orders. Switch only if the update performs better after slower deliveries have had time to arrive.

Potential expansion

- Peak-season guard: keep promise buffers wider going into Nov–Dec
- Carrier tracking data: update a parcel's risk while it is in transit
- Pilot with a control group: measure how many late parcels and bad reviews rescues prevent

LIVE DEMO

One morning screen: see → ask → act → check



